

Raju Kumar Paswan

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Arlington, TX

EDUCATION

The University of Texas at Arlington

Arlington, TX

Bachelor of Science in Computer Science; Minor in Data Science; GPA: 3.46/4.00

Expected May 2028

- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Computer Organization, Operating Systems, Discrete Mathematics, Software Engineering
- **Organizations:** ACM Research • ACM • Data Science Club • CSEC • HackUTA

EXPERIENCE

Student Technical Assistant - Academic Technology, UTA OIT

Arlington, TX

Office of Information Technology

Sep. 2026–Present

- Troubleshoot classroom and lab technology across hardware, software, AV systems, printers, and peripherals; document issues and escalate complex cases using structured problem-solving.

PROJECTS

OperatorLoop - Human-in-the-Loop Manufacturing AI — *Python, Pandas, scikit-learn, Matplotlib, Git* Aug-Sep 2026

- Built a 5-stage Python decision-support pipeline spanning sensor-data processing, anomaly classification, guidance retrieval, safety validation, and operator-feedback logging.
- Eliminated mismatched primary recommendations during testing by prioritizing exact process-condition matches ahead of TF-IDF similarity ranking.
- Evaluated retrieval and safety-validation behavior across 160 synthetic runs with 28 injected anomalies (17.5%); refined validation logic to reduce false-positive constraint violations.
- Created vibration-monitoring visualizations and published reproducible code with technical documentation on GitHub.

AArch64 Teaching Kernel Lab — *C, ARM Assembly, QEMU, GDB, Make, Git*

Sep. 2026

- Built and inspected a bare-metal 64-bit ARM kernel image for Raspberry Pi 3/QEMU; analyzed ELF sections and symbols with nm, objdump, and binary inspection tools.
- Validated the boot flow in QEMU/GDB by breaking at `_start`, stepping through AArch64 instructions, and inspecting register and exception-level state.
- Established a reproducible build/debug workflow with Make, shell scripts, and Git/GitHub to track code changes and document low-level debugging results.

MavRAG Course Assistant — *Python, FastAPI, LangChain, ChromaDB, Docker, AWS*

Expected Dec. 2026

- Planned RAG architecture to ingest course PDFs, generate embeddings, store vectors, and retrieve grounded context with source citations for LLM-based question answering.
- Planned REST APIs, semantic-search evaluation, prompt/version tracking, caching, Docker deployment, AWS EC2/S3 hosting, retrieval tests, and latency/error logging.

Cloud-Native Task Platform — *Java, Spring Boot, React, MySQL, Redis, AWS*

Expected Jan. 2027

- Planned three-tier system with a React client, RESTful Spring Boot API, MySQL data layer, JWT authentication, validation, pagination, indexing, and role-based access control.
- Planned Redis caching, Docker, load testing, GitHub Actions CI/CD, AWS RDS/EC2 deployment, CloudWatch monitoring, unit/integration tests, and horizontal scaling.

CERTIFICATIONS

AWS Certified AI Practitioner

Expected Oct. 6, 2026

AWS Certified Cloud Practitioner

Expected Oct. 15, 2026

TECHNICAL SKILLS

Languages: Java, Python, C, C++, SQL, JavaScript, HTML/CSS

Databases: MySQL, SQL, relational modeling; currently learning Redis, vector databases, DynamoDB, and NoSQL concepts

Verified Tools/Libraries: Git, GitHub, Linux, Pandas, scikit-learn, Matplotlib, TF-IDF, QEMU, GDB, Make

Cloud/DevOps - Currently Learning: AWS EC2, S3, RDS, Lambda, IAM, API Gateway, CloudWatch; Docker, GitHub Actions, CI/CD, infrastructure fundamentals

Frameworks - Currently Learning: FastAPI, React, Spring Boot, LangChain; RESTful API design, authentication, dependency management, API documentation

AI/ML - Currently Learning: Generative AI, LLMs, RAG pipelines, embeddings, semantic search, vector retrieval, prompt engineering, model evaluation

Core Concepts: Data structures and algorithms, object-oriented design, software design patterns, three-tier architecture, cloud-native systems, distributed systems, microservices, scalability, caching, testing, debugging, monitoring, SDLC, Agile development